

Network Calculus p. 2

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Network Calculus

- A theory for the determination of backlogs and delays in networks of network elements.
- Network elements comprise processors, servers, switches,...
- Network elements are modeled as buffer and service elements



- Based on input- and server-characteristic backlog- and delay-bounds as well as output characteristics are computed
- All models are lossless (no data lost and none produced)

Arrival

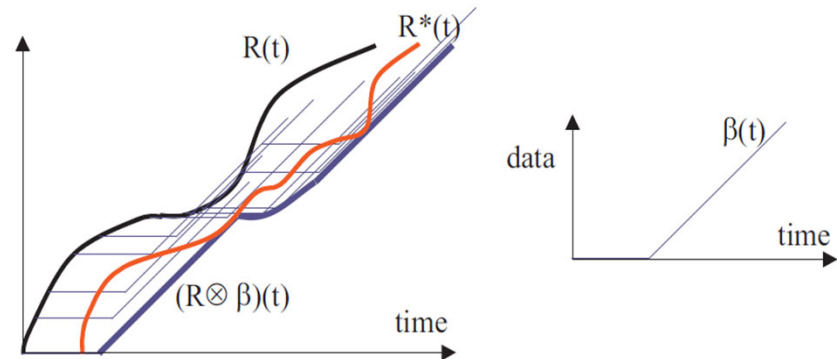
- Accumulated arrival functions $R(t)$: traffic received in $[0, t]$
- Arrival function may be constrained: $R(t) - R(s) \leq \alpha(t-s)$
- $\alpha(t-s)$ is typically either affine or staircase
- Packet spacing with period T and tolerance τ is constrained by staircase arrival curve – covers the periodic case (and periodic with jitter)
- For fixed length packets affine and staircase constraints are equivalent.
- Leaky buckets and GCRA's are equivalent and are affine and staircase constrained.

Service curves

- Service curves specify in the worst service to be experienced by some service element in a network.
- Service curves play the role of transfer functions (system impulse responses in our network calculus)

DEFINITION 1.3.1 (Service Curve). *Consider a system S and a flow through S with input and output function R and R^* . We say that S offers to the flow a service curve σ if and only if σ is wide sense increasing, $\sigma(0) = 0$ and $R^* \geq R \otimes \sigma$, i.e*

$$(R(t) \geq) \quad R^*(t) \geq \inf_{s \leq t} (R(s) + \sigma(t - s))$$



Service curves (motivation)

- Consider a service element where the output rate is constantly r within busy periods.
Input flow is R and output flow is R^*

- We have directly for $s < t$

- $R^*(t) - R^*(s) = r(t-s)$

- when t and s are in the same busy period.

- If s is the beginning of the busy period:

- $R(s) = R^*(s)$

- so that

$$R^*(t) - R(s) = r(t-s) \Rightarrow R^*(t) = R(s) + r(t-s)$$

- and in turn

$$R^*(t) \geq \inf_{0 \leq s \leq t} \{R(s) + r(t-s)\}$$

Service Curves (motivation)

- Consider a service element where every bit is delayed constantly d .
- We have directly for $d < t$

$$R^*(t) = R(t-d)$$

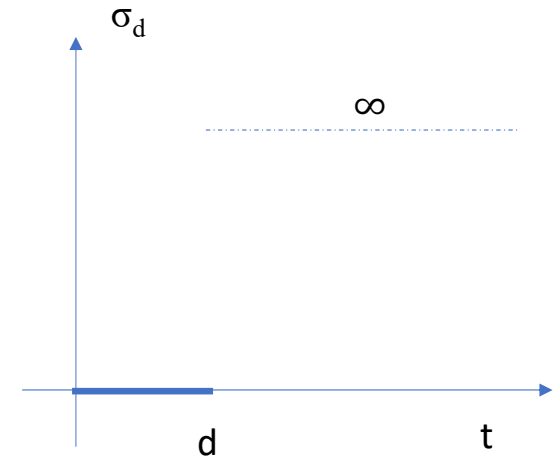
- Consider the function σ_d defined by

$$\sigma_d(t) = 0 \text{ for } t \leq d$$

$$\sigma_d(t) = \infty \text{ for } t > d$$

- Then

- $R^*(t) = R(t-d) = \inf_{0 \leq s \leq t} \{R(s) + \sigma_d(t-s)\}$
- (s must be the smallest value for which $t-s \leq d$ i.e. $s = t-d$)



Service curves (motivation) (strict service curves)

- Consider a network element where

$$R^*(t) - R^*(s) \geq \sigma(t-s) \Rightarrow R^*(t) \geq R^*(s) + \sigma(t-s)$$

When s and t are in the same busy period

- Then for s start of busy period where $R^*(s) = R(s)$

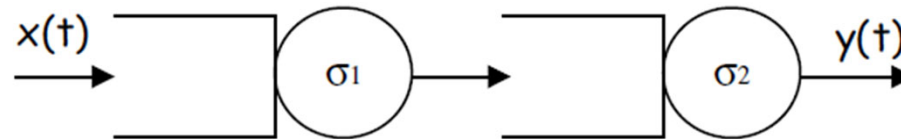
$$R^*(t) \geq R(s) + \sigma(t-s) \geq \inf_{0 \leq s \leq t} \{\sigma(t-s) + R(s)\}$$

- For some t in an idle period:

$$R^*(t) = R(t) \geq \inf_{0 \leq s \leq t} \{\sigma(t-s) + R(s)\}$$

Concatenation

THEOREM 1.4.6 (Concatenation of Nodes). *Assume a flow traverses systems $S1$ and $S2$ in sequence. Assume that S_i offers a service curve of σ_i , $i = 1, 2$ to the flow. Then the concatenation of the two systems offers a service curve of $\sigma_1 \otimes \sigma_2$ to the flow.*



PROOF:

- Call $R1$ the output of node 1, which is also the input to node 2.
- The service curve property at node 1 gives
- $R1 \geq R \otimes \sigma_1$
- and at node 2
- $R^* \geq R1 \otimes \sigma_2 \geq (R \otimes \sigma_1) \otimes \sigma_2 = R \otimes (\sigma_1 \otimes \sigma_2)$

Concatenating Constant Rates

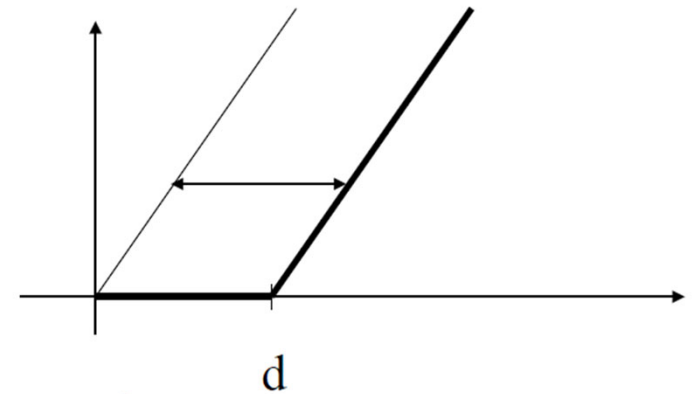
$$\begin{aligned}\sigma_1 \otimes \sigma_2 &= \inf_{0 \leq s \leq t} \{ \sigma_2(t-s) + \sigma_1(s) \} \\ &= \inf_{0 \leq s \leq t} \{ r_2(t-s) + r_1 s \} \\ &= \inf_{0 \leq s \leq t} \{ r_2 t + (r_1 - r_2) s \} \\ &= r_2 t \text{ for } r_1 > r_2 \quad (s=0) \\ &\quad r_1 t \text{ for } r_2 > r_1 \quad (s=t) \\ &= \min\{r_1, r_2\} t\end{aligned}$$

CONSISTENT WITH BOTTLENECK INTUITION

Concatenating with maximum delay

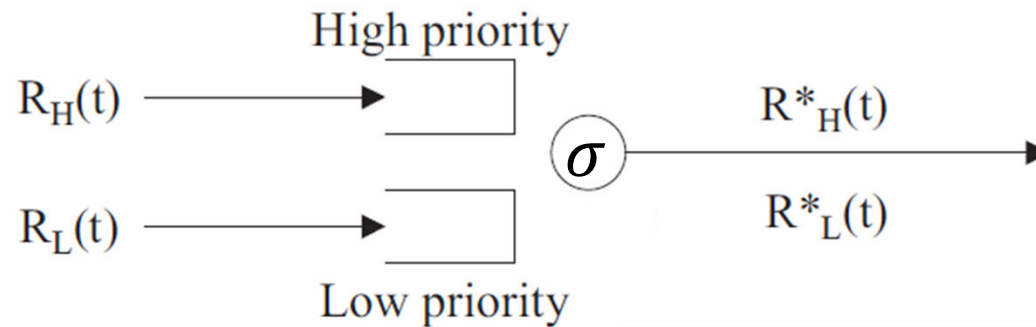
$$\begin{aligned}\sigma d \otimes \sigma &= \inf_{0 \leq s \leq t} \{ \sigma(t-s) + \sigma d(s) \} \\ &= \min \{ \inf_{s \leq d} \{ \sigma(t-s) \}, \infty \} \\ &= \inf_{s \leq d} \{ \sigma(t-s) \} = \sigma(t-d)\end{aligned}$$

or in fact $[\sigma(t-d)]^+$



Rate – latency curve

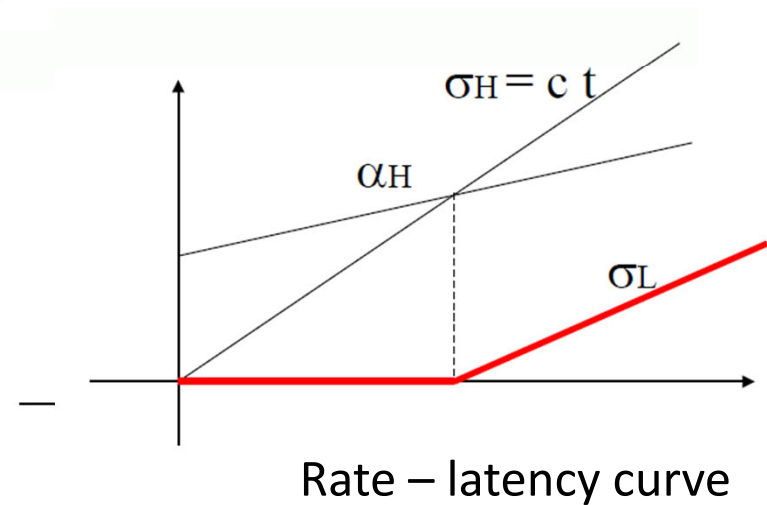
Prioritizing Servers (generally)



σ : strict

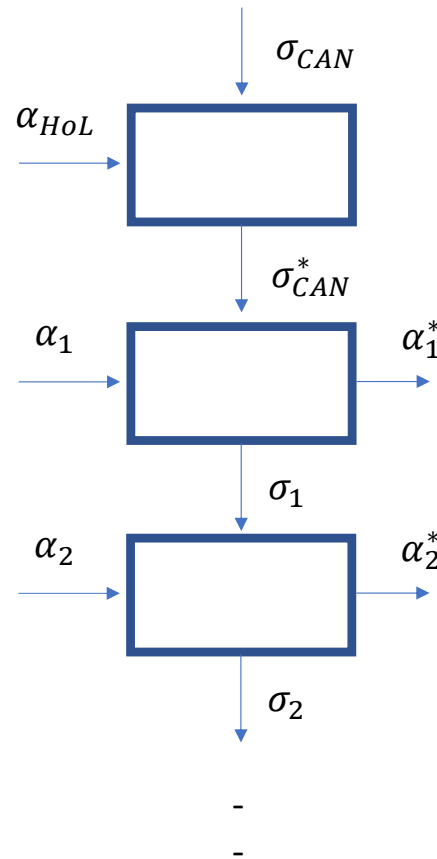
$$\sigma_H = \sigma$$

$$\sigma_L = [\sigma - \alpha_H]^+$$

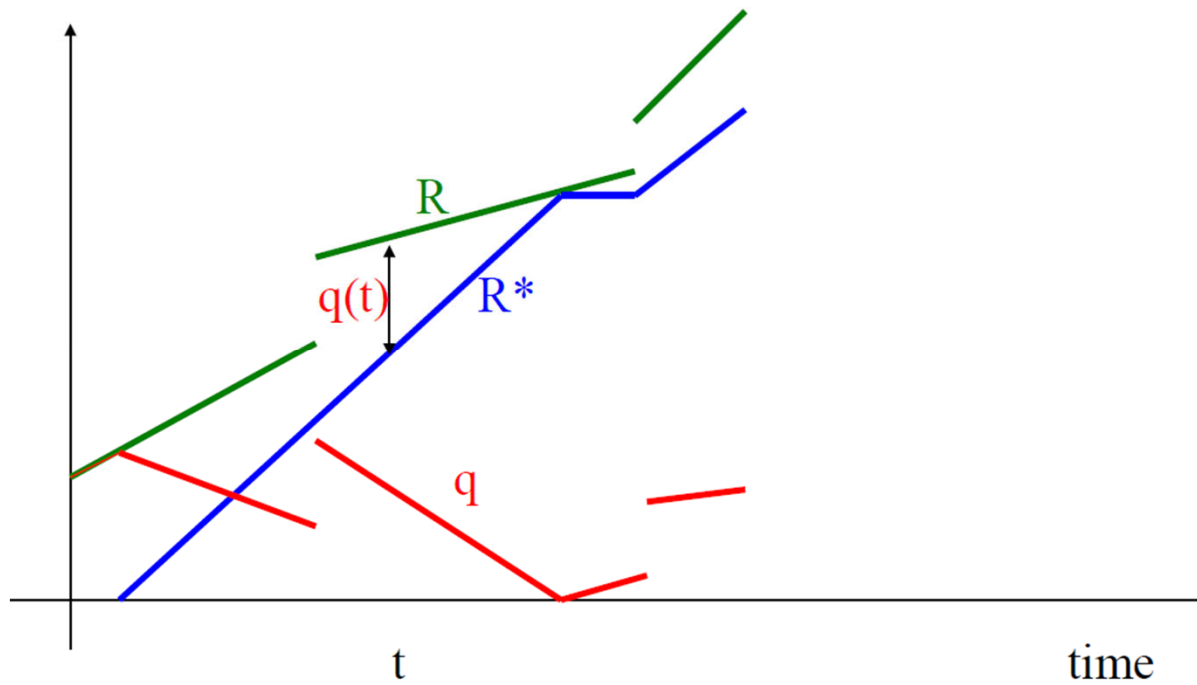


Prioritizing Servers (non-preemptive - CAN)

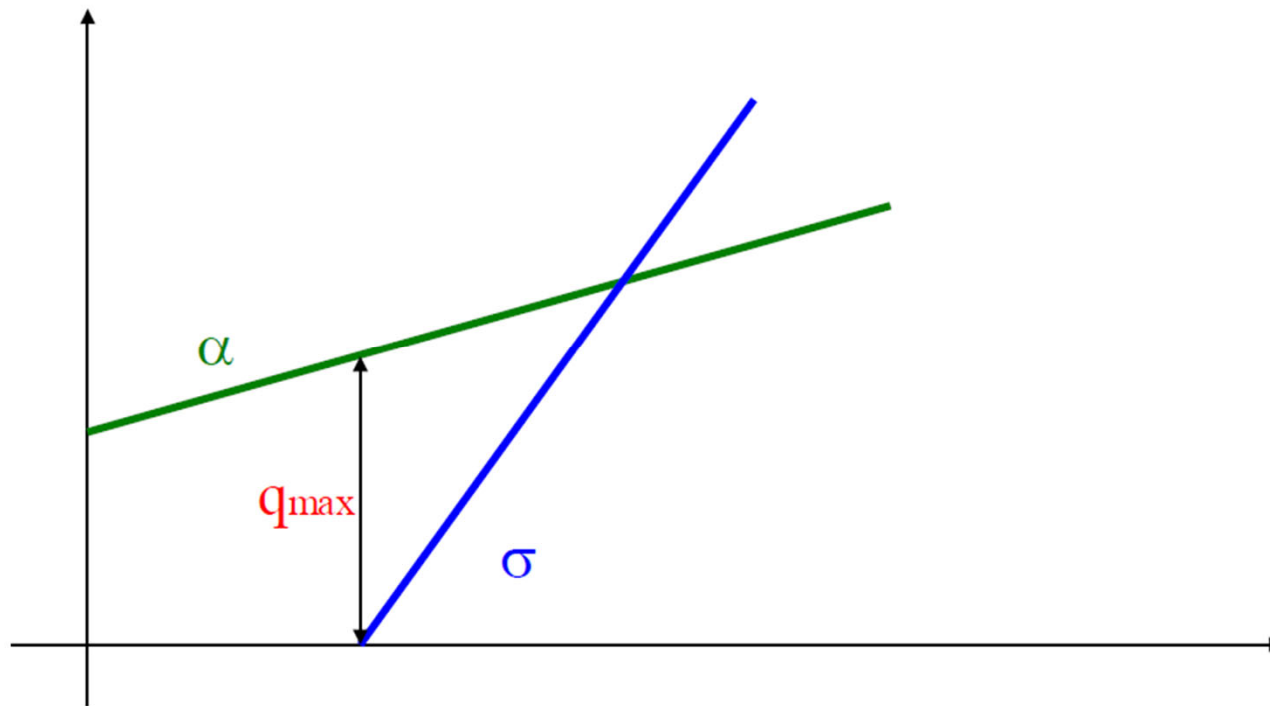
- $\sigma_{CAN} = r t$ (strict)
- $\alpha_{HoL} = k$ (Head of Line)



Backlog



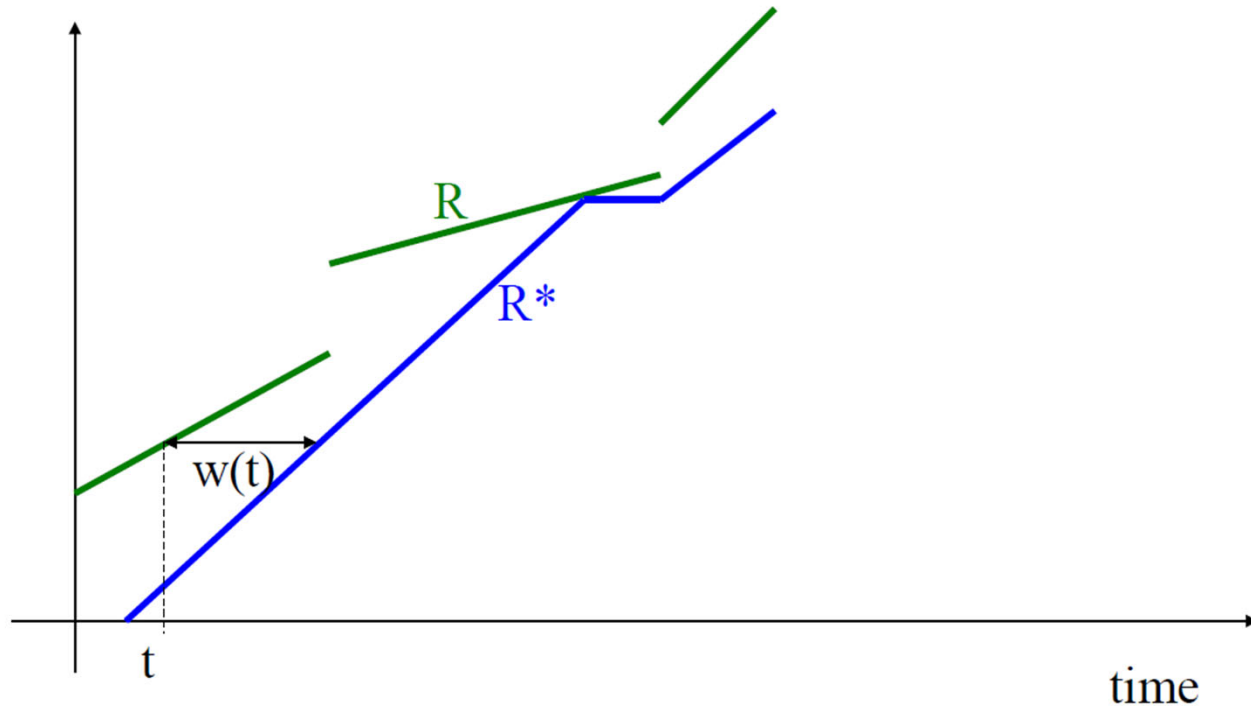
Backlog



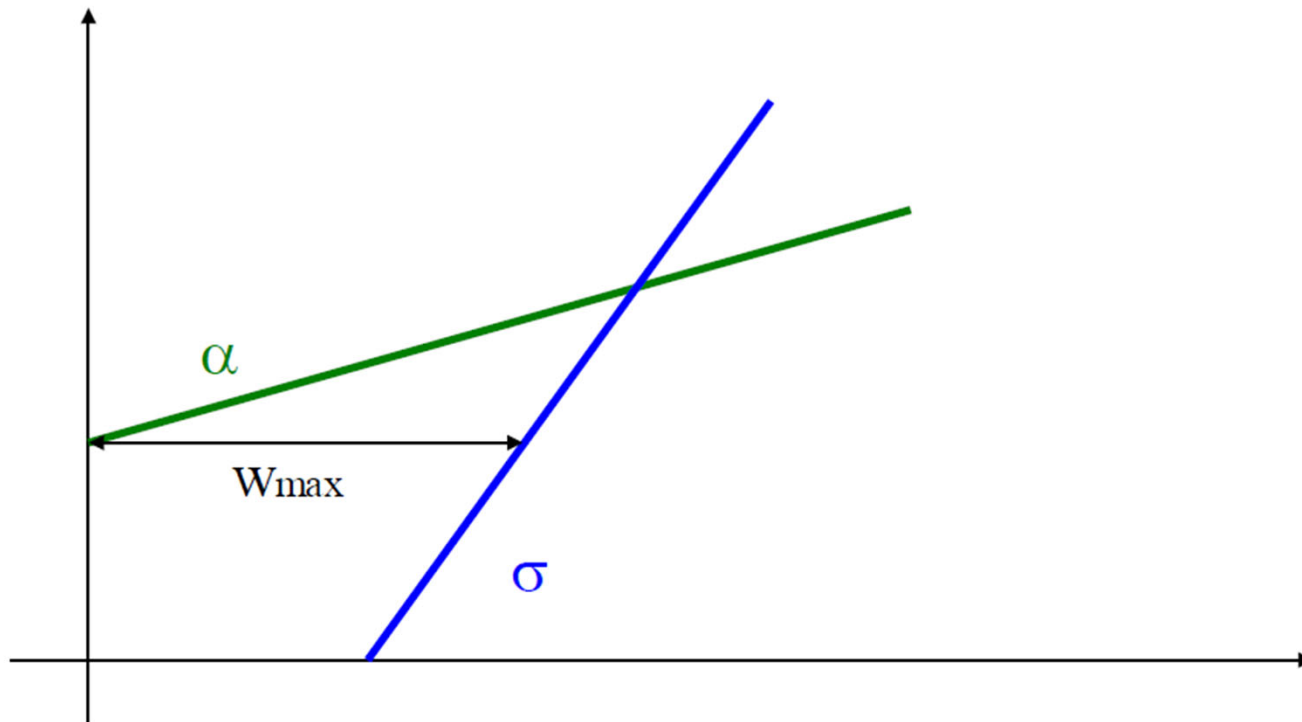
Proof

$$\begin{aligned} R(t) - R^*(t) &\leq R(t) - \inf_{s \leq t} \{ \sigma(t-s) + R(s) \} \\ &= \sup_{s \leq t} \{ R(t) - \sigma(t-s) - R(s) \} \\ &= \sup_{s \leq t} \{ R(t) - R(s) - \sigma(t-s) \} \\ &\leq \sup_{s \leq t} \{ \alpha(t-s) - \sigma(t-s) \} \\ &= \sup_{0 \leq t-s} \{ \alpha(t-s) - \sigma(t-s) \} \end{aligned}$$

Waiting times



Waiting time



Output Flow

- Let the inflow R to some service element be constrained by α and the service element guarantee a service curve σ
- Then the output flow R^* is constrained by

$$A^* = \alpha \oslash \sigma = \sup_{u \geq 0} \{ \alpha(t+u) - \sigma(u) \}$$

Output flow - example

$$\begin{aligned}
 \sup_{u \geq 0} \{ \alpha(t+u) - \sigma(u) \} &= \sup_{u \geq 0} \{ r(t+u) + b - [s u - p]^+ \} \\
 &= \max \{ \sup_{0 \leq u < d} \{ r(t+u) + b \}, \sup_{u \geq d} \{ r(t+u) + b - s u + p \} \} \\
 &= \max \{ \{ r(t+d) + b \}, \{ r(t+d) - (s d - p) \} \} \\
 &= r(t+d) + b = r t + r d + b = r t + r (p/s) + b
 \end{aligned}$$

